A Web-based Parcel Management System for a Courier Company

Project Outline

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Project Description

This project involves the design and implementation of a web-based Parcel Management System for a courier company operating within a configurable geographical area. The system will support both customers arranging parcel deliveries and internal staff responsible for managing orders and updating delivery progress. The aim is to create a realistic but simplified courier platform that reflects modern parcel services while remaining achievable within the scope of a single-developer final year project.

Customers will be able to place delivery orders either as registered users or as guests. Guest users will be allowed to complete orders and track parcels using a reference number, while registered users will benefit from additional features such as saved personal details and access to order history. Orders will involve arranging parcel collection from a specified address and delivery to a destination address. Both collection and delivery locations must fall within the company's defined operating area, which will be designed to allow expansion or reduction over time.

Each parcel will progress through several delivery states such as order placed, awaiting collection, collected, in transit, out for delivery, and delivered. The website will display detailed tracking information, including timestamps and location history presented in a user-friendly format, while SMS notifications will be limited to key events such as dispatch and delivery. This approach aims to keep users informed without overwhelming them with unnecessary updates.

An internal admin and staff interface will also be developed. This will allow authorised users to update parcel statuses, manage website content, and control limited-time offers or announcements displayed on the front page. This simulates real operational maintenance while keeping implementation complexity manageable.

The system will include online payment functionality using a third-party provider. Multiple payment options will be researched and compared before selecting the most suitable solution for the project, with payment initially implemented in a simulated or test environment. Development will be carried out using PHP and SQL within the University hosting environment, with supporting frontend technologies.

Proposed Tasks

The project will begin with requirements analysis and system design, defining user roles, parcel lifecycle states, database structure, and application architecture. Particular attention will be paid to balancing realism with feasibility and ensuring that both customer-facing and administrative features are clearly specified.

Research will then be conducted into external services required for the project, including payment providers and SMS gateways. A comparative analysis of available payment systems will be carried out, focusing on suitability, ease of integration, deployment constraints, and cost. A similar evaluation will be performed for SMS services before selecting appropriate technologies.

The main development phase will focus on implementing the core web application. This includes customer order placement, delivery area validation, parcel tracking, and payment processing, alongside the internal admin dashboard for managing orders, updating statuses, and editing website content such as promotions or notices.

Security considerations will also be addressed, particularly around guest usage. This will include preventing abandoned or automated order creation through measures such as rate limiting, input validation, and cleanup of incomplete transactions.

Further tasks include implementing SMS notifications for key delivery events, testing system functionality and usability, and refining the interface based on feedback. Regular supervisor meetings will be used to review progress and guide development decisions.

Project Deliverables

The project is expected to produce a working web-based Parcel Management System hosted within the University environment. This will include a customer-facing website allowing users to place delivery orders, make payments, and track parcels, alongside an internal admin and staff interface for managing orders, updating parcel statuses, and maintaining website content such as announcements or limited-time offers.

Technical deliverables will include the complete application source code (HTML, CSS, JavaScript, PHP, and SQL), a relational database schema supporting users, orders, parcels, and tracking history, and documented system architecture and design decisions.

A comparative analysis of online payment providers will be produced, leading to the selection and implementation of a suitable payment solution in test or simulation mode. A similar investigation will be carried out for SMS notification services, followed by integration for key delivery updates.

Additional deliverables include requirements documentation, security considerations for guest users and order handling, and testing evidence demonstrating system functionality and usability. The project will also produce a mid-project demonstration, a final system demonstration, and a comprehensive written report describing development, evaluation, and reflection on the project outcomes.

Initial Annotated Bibliography

https://teaching.dcs.aber.ac.uk/dates/Module/Index/2026/CS39440 - Module Management Platform (MMP) website used for project deadlines, assessment guidance, and module requirements.

https://www.php.net/docs.php - Official PHP documentation, intended to support backend development including form handling, sessions, and database interaction.

https://www.mysql.com/ - Reference site for MySQL database technology, relevant for designing and implementing the project's relational database.

https://www.royalmail.com - Royal Mail courier website, to be analysed for parcel tracking features and general user experience.

https://www.evri.com - Evri delivery service website, to be reviewed as part of a comparative study of courier platforms.

https://www.dpd.co.uk - DPD courier website, to be examined for tracking workflows and interface design ideas.

https://stripe.com/docs - Stripe documentation, included as a potential payment provider for researching test-mode payment integration.

Document Change History

Version	Date	Changes made to the document	Changed by
1.0	02/02/2026	Creation of document.	ard38
1.1	06/02/2026	Rewritten version in more details and reduced redundancy according to the supervisor’s (emi3) feedback	ard38